The Saty Phase Oscillator Opening-Window Strategy: Original, Replication and Improvements

An X quote-chain between The Milk Man (@MrMilkTrading) and the Dutch Warren (@etbent1), 26–27 August 2026

1. The original strategy — @MrMilkTrading, 26 Aug 2026

https://x.com/MrMilkTrading/status/2092599219521507760

> Ok let's talk about this strategy. The rules:
>
> Saty Phase Oscillator, NQ 10m chart (ETH)
>
> 1. PO closes above +100 --> long. Closes below -100 --> short
> 2. Entries 9:30-11:00 ET only
> 3. Exit on the opposite cross, or flat at the close
> 4. No stop, no target
>
> Since 2019, 1 NQ contract, friction included:
>
> 392 trades
> 59% win rate
> 1.86 profit factor
> +$259,883
> Max drawdown -$29,707

Its own quoted teaser post:

> One of the best backtests I have found this year
>
> The indicator is free, and half of you already have it on your chart
>
> It just printed a short on NQ tonight btw (outside my trading window, but go look)
>
> Any guesses?

2. The independent replication — @etbent1, 26 Aug 2026

https://x.com/etbent1/status/2092707152251191659

> Took you up on the "test it yourself."
>
> Rebuilt it from the Pine source — an MQL5 EA and a Python twin, neither tuned toward your numbers. Ran it on USTEC instead of NQ.
>
> You: 12,994 pts (2019–26)
> Me: 13,038 pts (2020–26, where my data starts)
>
> Different index, different window, ~2% apart. It replicates.
>
> First thing I checked was whether it's just beta, since the index rose the whole time. Buying every open in the same 9:30–11:00 window gives +6.5 pts/trade over 1,716 trades. Your signal: +35.0 over 373. It also beats 1000 out of 1000 random-entry variants. The edge is real.
>
> Two changes improved it, and both held on train (2020–23) and test (2024–26):
>
> 1. End the window at 10:00 instead of 11:00. Profit factor 1.97 → 2.34, and return/drawdown roughly doubles. It's monotone — every widening of the window makes it worse. That fits the construction: ATR(14) on a 10m chart is 140 minutes, so at 9:30 the normalizer is all premarket, and by 11:00 it's full of RTH bars. The 3-ATR threshold isn't measuring the same thing anymore.
>
> 2. Exit at 61.8 rather than the full traverse to −100. Profit factor 1.97 → 2.16. Saty already plots that level.
>
> Your threshold is right where it should be, though — I swept 80 through 120 and 100 is a clean peak on both halves, not a fitted edge.
>
> On the missing stop, I measured instead of guessing. Winners' MAE: median 35 pts, p95 137. Losers: median 117. Any stop tighter than 150 costs real money — at 60 pts you give up 53% of total profit, at 100 pts 21%. At 200 pts it costs 1%. So there's room for a catastrophe stop, but not for a working one.
>
> One caveat worth flagging: US500 gives the same answer, but only in the back half. Its 2020–23 profit factor is 1.14 against USTEC's 1.60.

3. The author's response — @MrMilkTrading, 26 Aug 2026

https://x.com/MrMilkTrading/status/2092710877845655590

> Love it when people feed my stuff to Claude and it not only replicates, but generates new ideas
>
> Kind of like outsourcing my token usage
>
> Thank you @etbent1

4. The VWAP-distance filter — @etbent1, 27 Aug 2026

https://x.com/etbent1/status/2092860879712801203

> One filter — skip entries sitting more than 2.5 ATR from session VWAP — ends in the same place on 258 trades instead of 373, with equity drawdown of 759 points instead of 1,797.

5. The filter confirmed on NQ — @MrMilkTrading, 28 Aug 2026

https://x.com/MrMilkTrading/status/2093162557536141453

> @etbent1 nailed it.
>
> Added his VWAP filter to the NQ Phase Oscillator setup: skip the entry if price is more than 2.5 ATR from VWAP.
>
> Same total money, lower drawdown. The trades it cut were basically coin flips.
>
> Re-deploying this in NinjaTrader -- thanks amigo
>
> his hint:

Attached chart — "PO Phase Oscillator on NQ, 10-min -- now with a VWAP filter / +100/-100 cross, 9:30-10:00 ET, exit at opposite 61.8 -- skip if price is > 2.5 ATR from VWAP". Before → after the VWAP filter:

Metric	Before	After
Profit factor	2.73	3.07
Max drawdown	-$14.0K	-$10.4K
Return / DD	16.1x	21.8x
t-stat	4.93	5.18

Win rate	64.3%	65.9%
Trades / yr	26	24
Total (1 NQ)	+$227K	+$227K
2026 YTD	+$53.6K	+$55.1K

Reply to @MarginCall4 asking "Low drawdown levered tech beta. How large of a sample size?":

> 182 trades since 2019, about 24 a year.
>
> Not really beta though -- shorts actually made more than longs ($140k vs $87k on 1 contract).
>
> Per trade it also beats buying every open by ~5x, and it beat 1000 out of 1000 random entry variants in the same window.